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birdweatheR: An R package for accessing and analyzing BirdWeather acoustic and environmental data

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Software

- Review
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Summary

The BirdWeather participatory science project provides a global database on avian vocal detections paired with environmental data, enabling new lines of research, but has been underrepresented in ecological research. We developed the `birdweatheR` R package, a streamlined interface to the BirdWeather GraphQL API, supporting reproducible ecological research within the R environment. We provide three case studies demonstrating how the package facilitates access and use of bird and environmental data for ecological analyses. BirdWeather holds considerable promise as a data source for behavioral ecology and macroecology, and birdweatheR will help fulfill this promise by making the database usable by a broad community of ecologists.

Statement of need

Passive acoustic monitoring combined with automated species identification is emerging as a powerful tool for broadscale biodiversity monitoring (Gibb et al., 2019; Kahl et al., 2021; Kershenbaum et al., 2025; Ross et al., 2023). While participatory-science projects such as eBird (Sullivan et al., 2009) have transformed our understanding of the distribution and abundance of biodiversity, acoustic monitoring offers an opportunity to move beyond the occurrence record and capture information on animal behavior at unprecedented scales. Species vocalizations encode information about behavior and environmental context, and today's autonomous acoustic networks operate continuously, facilitating advances in biodiversity monitoring and "macrobehavior" (Keith et al., 2023).

Established in 2021, BirdWeather (BirdWeather, 2024) is a global network of volunteer-deployed acoustic sensors with on-board species identification powered by the BirdNET algorithm (Kahl et al., 2021). As of early 2026, the network comprised nearly 18,000 recording stations across six continents that collect tens of millions of detections every month. While BirdWeather offers several recording options, many are Portable Universe Codecs (PUCs) that simutaneously record bird vocalization detections and temperature, humidity, barometric pressure, air quality, and spectral light levels at high temporal resolution via onboard sensors, enabling direct linkage of vocalization activity to local abiotic conditions.

The scientific potential of BirdWeather is substantial. Using BirdWeather detections, Pease & Gilbert (2025) demonstrated that bird vocalization activity is prolonged by nearly an hour in light-polluted areas, and Gilbert et al. (2026) used the platform to study the impact of solar eclipses on wildlife activity. Despite this potential, uptake among ecologists has been slow, likely due to API barriers and the absence of an analytical pipeline in R, the primary analytical environment for most ecologists (Lai et al., 2019). BirdWeather currently offers three data access modes: a graphical user interface, a REST API, and a GraphQL API—an effort in data sharing that exceeds most comparable platforms, but one that has yet to translate into broad scientific use. `birdweatheR` solves this gap by targeting ecologists and ornithologists who are familiar with R but lack the programming background to interact directly with GraphQL or REST APIs.

State of the field

Several R packages exist for accessing biodiversity occurrence data, such as `rebird` for eBird (Maia et al., 2021) and `rgbif` for GBIF (Chamberlain & Boettiger, 2017), or even `sppocc` for downloading species occurrence data from several databases simultaneously (**?**). There are additionally a number of packages developed for analyzing raw acoustic recordings, such as `warbleR` (Araya-Salas & Smith-Vidaurre, 2017), `monitoR` (Katz et al., 2016), and seewave (J. Sueur et al., 2008). `birdweatheR` differs from the previously mentioned packages in that it provides programmatic access to a continuously operating autonomous acoustic detection network that pairs species-level vocalizations with concurrent environmental sensor readings - a combination of data currently unavailable in R. So, although a number of packages exist for accessing ecologically relevant biodiversity data, no database mirrors BirdWeather, emphasizing the ecological and scientific importance of `birdweatheR`.

Software design

`birdweatheR` is an R package that provides a streamlined interface to the BirdWeather GraphQL API (BirdWeather, 2024), enabling ecologists to access avian vocalization detections, summarize species activity patterns, and retrieve environmental sensor data without direct interaction with the underlying API infrastructure (Figure 1).

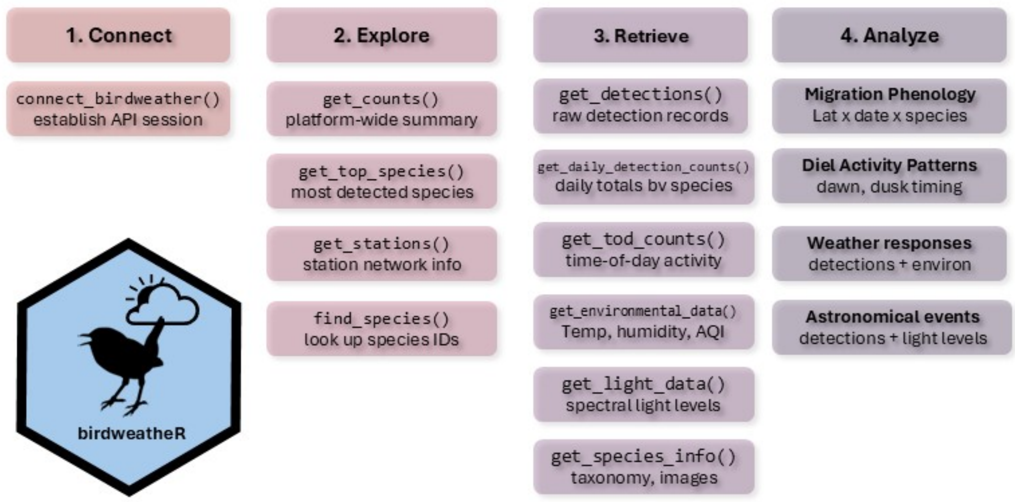

Figure 1: Overview of the `birdweatheR` package. The package stores an (1) API connection with no required credentials, (2) provides platform exploration functionality, (3) enables retrieval of the database, with (4) a number of ecological applications possible.

A single function call can retrieve thousands to millions of records and return them as a clean, analysis-ready data frame. Package functionality spans platform-wide and regional detection summaries (`get_counts`, `get_top_species`), daily and time-of-day activity patterns for diel niche analyses (`get_daily_detection_counts`, `get_tod_counts`), raw detection downloads with flexible filtering by species, date range, and geographic bounding box (`get_detections`), and retrieval of environmental and spectral light sensor time series (`get_environment_data`, get_light_data) Table 1.

Function	Description
connect_birdweather()	Establishes a connection to the BirdWeather GraphQL API. Must be called once per session before using any other package functions. Stores the connection in a package-level environment accessible to all downstream functions.
find_species()	Searches the BirdWeather species database by common name, scientific name, or partial match (including wildcards). Returns a data.table with species IDs, common names, and scientific names. Useful for looking up species IDs to pass into other functions.
get_counts()	Returns a single-row summary of total detections, unique species, and active stations for a given time period. Supports filtering by station IDs, station types, species, and a geographic bounding box. Useful as a quick platform-wide or regional snapshot summary.
get_top_species()	Returns the most frequently detected species for a given time period, ranked by total detection count. Includes a confidence breakdown (almost certain, very likely, unlikely, uncertain) for each species.
get_daily_detection_counts()	Returns detection counts aggregated by day for a given time period. Can optionally return a species-level breakdown (one row per species per day). Supports filtering by station IDs and species IDs. Useful for visualizing seasonal trends without downloading raw detections.
get_detections()	Retrieves raw bird detections from the BirdWeather API. Returns a flat data.table with detection metadata (detection ID, timestamp, confidence, and probability) and expanded species and station fields. Supports filtering by date range, station, station type, species, continent, confidence threshold, probability, and geographic bounding box. Handles pagination automatically.
get_stations()	Retrieves public BirdWeather stations with optional filters. Returns a flat data.table with station ID, name, and type, continent, country, state, coordinates, and location privacy setting. Supports filtering by name, time period, and geographic bounding box. Handles pagination automatically.

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Function	Description
`get_tod_counts()`	Returns detection counts binned by 30-minute time-of-day intervals for a given species, revealing diel activity patterns. Supports filtering by date range, station, confidence threshold, bounding box, and time window. Can return per-station breakdowns and optionally fill zeros for undetected bins.

Table 1: Major functions of the `birdweatheR` package.

The `birdweatheR` R package is publicly available at: https://github.com/BrentPease1/birdweatheR. The package is designed to provide an R interface to the BirdWeather database, supporting use of BirdWeather in ecologists' preferred analytical software, promoting data analysis, and increasing scientific reproducibility (Lai et al., 2019). We developed package functionality for:

1. **Exploration**: `birdweatheR` provides platform-wide exploratory functions for users to identify species or spatiotemporal extents of the dataset. For example, users can query the platform with `get_counts` or `get_top_species` to learn about platform-wide or species-specific detection counts, respectively, during a given time period. Users can also use the `get_stations` function to identify stations on a continent or within a bounding box. The `find_species` function provides access to which species occur in the dataset and provides users with the platform-specific species identification number for filtering downstream analyses.

2. **Retrieve**: The core retrieval function in `birdweatheR` is `get_detections`. This function downloads raw bird vocalization detections from the BirdWeather API with flexible filtering options, returning a flat data.table with detection, species, and station information. The function handles pagination automatically, enabling users to download complete datasets from the platform. Users can also query temporal dynamics of vocalization activity through `get_tod_counts` and `get_daily_detection_counts`, both of which provide hourly or daily counts, respectively, with the latter function allowing for species-level information. We additionally provide functions for downloading environmental and spectral light time series data from BirdWeather PUCs with `get_enviroment_data` and `get_light_data`, respectively. Once PUC `station_ids` of interest are identified during the exploratory or data retrieval stage, a character vector of `station_ids` along with a time window can be provided to either function to retrieve environmental data at approximately one-minute intervals.

Several design decisions were made during development. Namely, a key package dependency of `data.table` (Barrett et al., 2026) was intentional due to its performance at scale, as the BirdWeather database passed 2 billion observations in February 2026. We also chose to use GraphQL API calls instead of REST API calls due to the flexibility of field selection reducing the payload size of each API call. Given that the Birdweather GraphQL API doesn't require credentials, we chose to use a package-level environment for the connection object; if the database changes, we'll provide updates to `birdweatheR` to allow for user credentials. We also had to make decisions about downstream cleaning. BirdWeather requires many analytical decisions such as BirdNET confidence thresholds (Wood et al., 2023) and stationary vs traveling stations. We decided to leave the dataset in its rawest form and let each user decide best practices for data cleaning and vetting.

Research impact statement

The `birdweatheR` package will reduce barriers for researchers to use the BirdWeather database for ecological analyses. The program, by providing detections at fine spatiotemporal resolutions from standardized sensors, provides a rich source to fuel investigations into the behaviors and distributions of the world's birds. We have already used BirdWeather to provide syntheses of how birds respond behaviorally to light pollution (Pease & Gilbert, 2025) and solar eclipses (Gilbert et al., 2026), and we are particularly hopeful that the dataset will advance the emerging field of macrobehavior (Keith et al., 2023). To reach its potential as a research tool, BirdWeather must be accessible and usable by the broadest audiences of ecologists as possible, and our package is a significant step in that direction. To further illustrate the research impact and potential of the package, we provide the following three worked examples, aimed at stimulating research from the scientific community.

Applications

We demonstrate birdweatheR through three worked examples that span the package's core analytical capabilities; fully reproducible code for each is provided in the package vignettes at https://brentpease1.github.io/birdweatheR/.

1. Migration phenology.

Because BirdWeather stations operate continuously, the dataset can be used for tracking the phenology of migratory species at continental scale. While this case study involves analysis at daily intervals, the most novel research opportunities provided by the database involves analyses at finer temporal resolutions (see Case Studies 2 & 3). Here we use the Wood Thrush (*Hylocichla mustelina*), a long-distance Neotropical migrant that winters in Central America and breeds across eastern North America, to illustrate how `get_detections()` can reveal migratory timing across latitudes. The `woth` dataset bundled with this package contains 569,000 Wood Thrush detections from North American BirdWeather stations during spring migration (March–May 2025), filtered to detections with a BirdNET confidence 0.6 (Wood et al., 2023).

We then calculated the median latitude of all detections for each day. As birds depart their Central American wintering grounds and move northward, the median latitude of detections shifts from roughly 15°N in early March to over 40°N by mid-May (Figure 2). This result reflects the known distribution and phenology of the species (Evans et al., 2020) and suggests that BirdWeather is a reliable source of high-resolution biodiversity data.

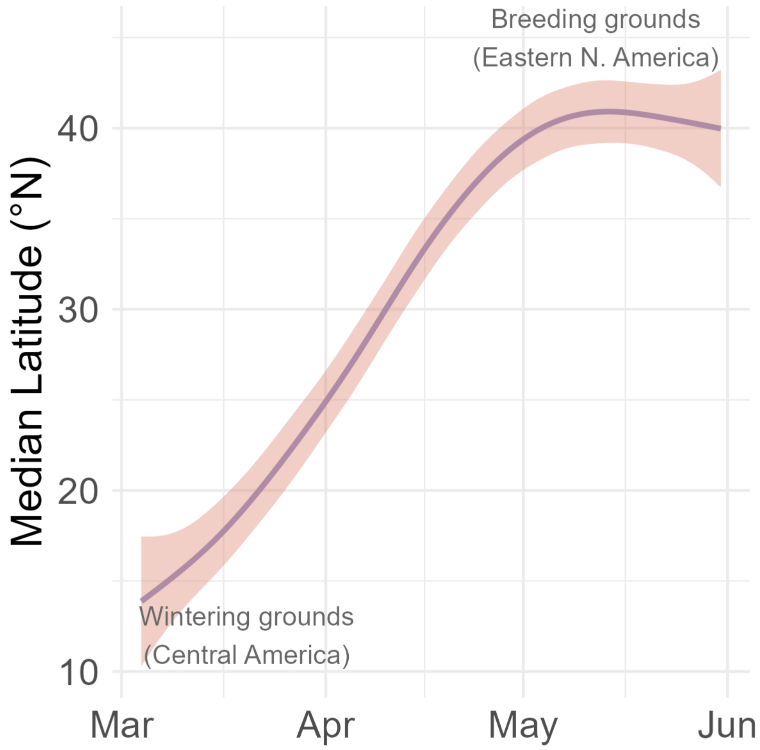

Figure 2: Wood Thrush (*Hylocichla mustelina*) spring migration phenology during March-May 2025.

2. Diel activity patterns.

While previous participatory-science programs like eBird have been used to track migration, BirdWeather is arguably the first program to enable quantification of avian activity patterns given the collection of detections with precise timestamps. Here, using the BirdWeather `woth` dataset, we quantify how wood thrush daily activity patterns vary between areas with differing levels of human disturbance (Kennedy et al., 2019). Using a trigonometric generalized linear mixed-effects model described by (Iannarilli et al. (2025)), we estimate diel patterns of vocalization for the Wood Thrush for March through May of 2025. The analysis reveals that thrushes in high-footprint landscapes begin vocalizing earlier in the morning than those in low-footprint areas (Figure 3), consistent with light-pollution-driven shifts in activity timing (Pease & Gilbert, 2025). This example illustrates how BirdWeather's continuous, timestamped detections enable diel niche analyses that are not possible with occurrence-based participatory science platforms.

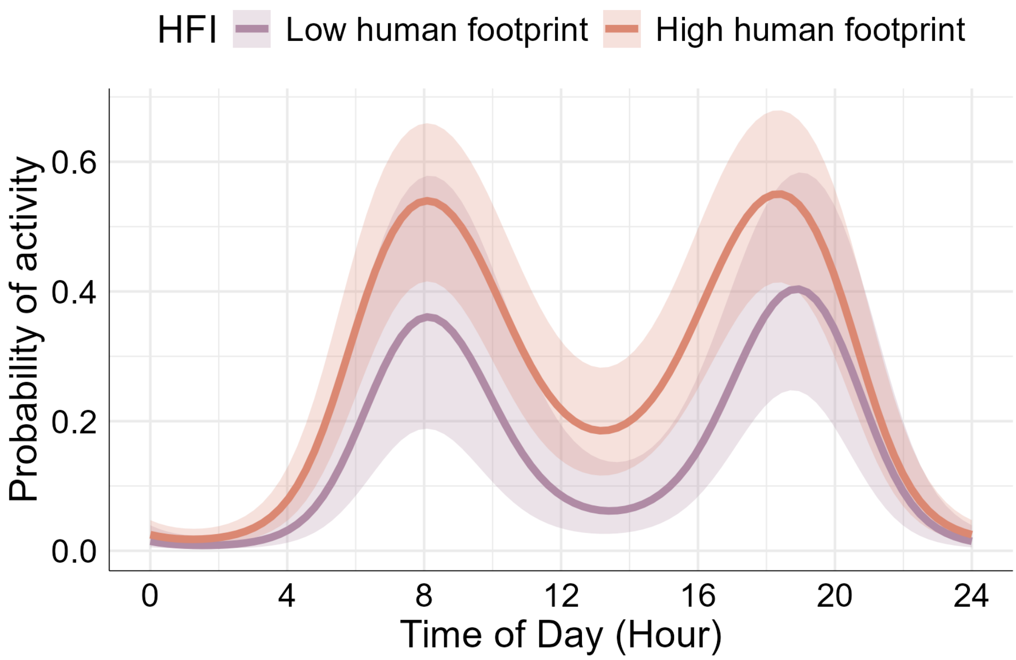

Figure 3: Wood Thrush (*Hylocichla mustelina*) diel activity patterns during March-May 2025 predicted for landscapes with low (purple) and high (orange) levels of human footprint.

3. Behavioral responses to an astronomical event.

BirdWeather PUCs record light levels approximately every minute, providing readings for broadband light levels, near-infrared light, and spectral bands F1–F8. The combination of acoustic detections and on-board light sensors makes BirdWeather data uniquely suited to studying how birds respond to rapid changes in light levels, including those caused by solar eclipses (Aguilar et al., 2025; Gilbert et al., 2026) . On April 8, 2024, a total solar eclipse crossed North America along a path running from Texas northeast through the Ohio Valley during the afternoon hours. We integrated `get_detections`, `get_light_data`, and `get_environment_data` to compile the `total_eclipse` dataset, which contains detections from PUC stations within an eclipse path bounding box during April 8, 2024. The `eclipse_light_data` and `eclipse_env_data` datasets contain concurrent light sensor and environmental readings from those same stations, respectively.

Bird vocalizations, PUC-recorded light levels, and humidity all declined markedly during the two-hour window surrounding totality (Figure 4), with humidity beginning to drop approximately 30 minutes before totality — consistent with known eclipse meteorology. This example highlights the unique value of BirdWeather's paired acoustic and environmental sensor data for studying behavioral responses to discrete environmental events.

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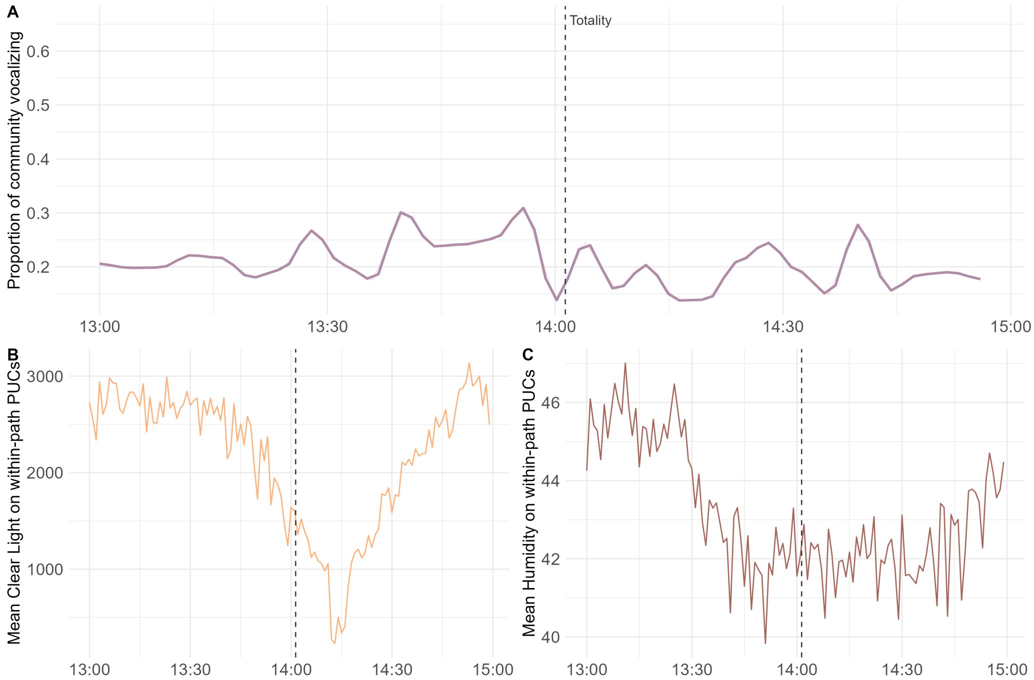

Figure 4: (A) Proportion of avian community vocalizing during 4-minute windows during April 8, 2024 between 13:00-15:00 CST. The total vocalizing community consists of total richness during 2 hour window. (B) Mean clear night levels from PUC sensors within the path of totality on April 8, 2024 during 13:00 – 15:00 CST. (C) Mean humidity levels within the path of totality on April 8, 2024 during 13:00 – 15:00 CST.

AI usage disclosure

Generative AI tools were used during the preparation of this work to assist with code development and testing. Claude Code (Anthropic, Claude Sonnet 4.6) was used to assist in API calls and documentation.The authors reviewed and verified all AI-generated code and take full responsibility for the accuracy and integrity of the final publication.

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References

Aguilar, L. A., Miller-Crews, I., Dobris, J. M., Tracy, J. A., Macklin, P., Dixit, S., Jacobson, R. A., Evans, R. L., McGuire, E. L., Beverly, D. P., Reichard, D. G., & Rosvall, K. A. (2025). Total solar eclipse triggers dawn behavior in birds: Insights from acoustic recordings and community science. *Science*, *390*(6769), 152–155. https://doi.org/10.1126/science.adx3025

Araya-Salas, M., & Smith-Vidaurre, G. (2017). warbleR: An r package to streamline analysis of animal acoustic signals. *Methods in Ecology and Evolution*. https://dx.doi.org/10.1111/2041-210X.12624

Barrett, T., Dowle, M., Srinivasan, A., Gorecki, J., Chirico, M., Hocking, T., Schwendinger, B., & Krylov, I. (2026). *Data.table: Extension of 'data.frame'*. https://doi.org/10.32614/

CRAN.package.data.table

BirdWeather. (2024). *BirdWeather: A living library of bird vocalizations*. https://www.birdweather.com/

Chamberlain, S., & Boettiger, C. (2017). R python, and ruby clients for GBIF species occurrence data. *PeerJ PrePrints*. https://doi.org/10.7287/peerj.preprints.3304v1

Evans, M., Gow, E., Roth, R., Johnson, M., & Underwood, T. (2020). Wood thrush (*hylocichla mustelina*), version 1.0. In *Birds of the world*. Cornell Lab of Ornithology. https://doi.org/10.2173/bow.woothr.01

Gibb, R., Browning, E., Glover-Kapfer, P., & Jones, K. E. (2019). Emerging opportunities and challenges for passive acoustics in ecological assessment and monitoring. *Methods in Ecology and Evolution*, *10*(2), 169–185. https://doi.org/10.1111/2041-210X.13101

Gilbert, N. A., Pease, B. S., Severino, M., & Winter III, H. T. (2026). Photic niche explains avian behavioral responses to solar eclipses. *Ecology and Evolution*, *16*(2), e73090. https://doi.org/10.1002/ece3.73090

Iannarilli, F., Gerber, B. D., Erb, J., & Fieberg, J. R. (2025). A "how-to" guide for estimating animal diel activity using hierarchical models. *Journal of Animal Ecology*, *94*(2), 182–194. https://doi.org/10.1111/1365-2656.14213

J. Sueur, T. Aubin, & C. Simonis. (2008). Seewave: A free modular tool for sound analysis and synthesis. *Bioacoustics*, *18*, 213–226. https://doi.org/10.1080/09524622.2008.9753600

Kahl, S., Wood, C. M., Eibl, M., & Klinck, H. (2021). BirdNET: A deep learning solution for avian diversity monitoring. *Ecological Informatics*, *61*, 101236. https://doi.org/10.1016/j.ecoinf.2021.101236

Katz, J., Hafner, S. D., & Donovan, T. (2016). Tools for automated acoustic monitoring within the r package monitoR. *Bioacoustics*, *25*(2), 197–210. https://doi.org/10.1080/09524622.2016.1138415

Keith, S. A., Drury, J. P., McGill, B. J., & Grether, G. F. (2023). Macrobehaviour: Behavioural variation across space, time, and taxa. *Trends in Ecology & Evolution*. https://doi.org/10.1016/j.tree.2023.08.007

Kennedy, C. M., Oakleaf, J. R., Theobald, D. M., Baruch-Mordo, S., & Kiesecker, J. (2019). Managing the middle: A shift in conservation priorities based on the global human modification gradient. *Global Change Biology*, *25*(3), 811–826. https://doi.org/10.1111/gcb.14549

Kershenbaum, A., Akçay, Ç., Babu-Saheer, L., Barnhill, A., Best, P., Cauzinille, J., Clink, D., Dassow, A., Dufourq, E., Growcott, J., Markham, A., Marti-Domken, B., Marxer, R., Muir, J., Reynolds, S., Root-Gutteridge, H., Sadhukhan, S., Schindler, L., Smith, B. R., & Dunn, J. C. (2025). Automatic detection for bioacoustic research: A practical guide from and for biologists and computer scientists. *Biological Reviews*, *100*(2), 620–646. https://doi.org/10.1111/brv.13155

Lai, J., Lortie, C. J., Muenchen, R. A., Yang, J., & Ma, K. (2019). Evaluating the popularity of r in ecology. *Ecosphere*, *10*(1), e02567. https://doi.org/10.1002/ecs2.2567

Maia, R., Chamberlain, S., Teucher, A., & Pardo, S. (2021). *Rebird: R client for the eBird database of bird observations*. https://doi.org/10.32614/CRAN.package.rebird

Pease, B. S., & Gilbert, N. A. (2025). Light pollution prolongs avian activity. *Science*, *389*(6762), 818–821. https://doi.org/10.1126/science.adv9472

Ross, S. R. P.-J., O'Connell, D. P., Deichmann, J. L., Desjonquères, C., Gasc, A., Phillips, J. N., Sethi, S. S., Wood, C. M., & Burivalova, Z. (2023). Passive acoustic monitoring provides a fresh perspective on fundamental ecological questions. *Functional Ecology*,

The Journal of Open Source Software

37(4), 959–975. https://doi.org/10.1111/1365-2435.14275

Sullivan, B. L., Wood, C. L., Iliff, M. J., Bonney, R. E., Fink, D., & Kelling, S. (2009). eBird: A citizen-based bird observation network in the biological sciences. *Biological Conservation*, *142*(10), 2282–2292. https://doi.org/10.1016/j.biocon.2009.05.006

Wood, C. M., Barceinas Cruz, A., & Kahl, S. (2023). Pairing a user-friendly machine-learning animal sound detector with passive acoustic surveys for occupancy modeling of an endangered primate. *American Journal of Primatology*, *85*(8), e23507. https://doi.org/10.1002/ajp.23507